

Vibhrav Jha

Madison, WI | Open to Relocate

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Junior at UW–Madison using applied AI and full-stack engineering to ship products that move business outcomes. At VDart, owned an automated RFI/RFP pipeline that cut proposal turnaround from 2 weeks to under 24 hours, tripled team proposal capacity, and now serves 50+ users in production. Currently building a Kotlin Android safety overlay app as part of an IRB-approved HCI study on real-time, on-device security guidance for adults with intellectual and developmental disabilities.

EXPERIENCE

UW–Madison People & Robots Lab

Madison, WI

Undergraduate Research Assistant

Jan 2026 – Present

- Building a **Kotlin Android** safety-guidance overlay app for adults with IDD as part of an IRB-approved HCI study, that analyzes on-screen activity using AccessibilityService and MediaProjection APIs, then routes structured prompts to Gemini for real-time risk and security guidance.
- Reduced unsafe model behavior by implementing a deterministic RiskAssessor layer that overrides low-risk model outputs for irreversible user actions.

VDart Inc.

Atlanta, GA

Software Engineering Intern

May 2025 – Aug 2025

- Cut proposal turnaround from **2 weeks to under 24 hours** and grew team capacity **3×** by owning an automated RFI/RFP pipeline built on **IBM Docling**, **Llama 3**, and **Gemini**, directly expanding the volume of deals the proposal team could pursue.
- Served **50+ users** in production by designing a **FastAPI** backend for model response parsing and orchestration, with a **React** chat interface featuring token streaming and drag-and-drop file upload.
- Hit **96% retrieval accuracy** against historical proposal benchmarks by building layout-aware document ingestion with semantic chunking that preserved tables, headers, and hierarchy across 100+ technical RFPs.

Pink City Expressways

Delhi, India

Software Development Intern

Jan 2022 – Mar 2023

- Grew successful partnerships from **15 to 19** and generated **\$18,000** additional revenue within 8 months by building a **Python + SQL** CRM tracking 30+ leads with normalized lead, contractor, and partnership records.
- Reduced client inquiry response times **40%** by shipping a responsive marketing website in **HTML**, **CSS**, and **JavaScript** and an automation tool matching machinery availability to demand.

PROJECTS

Distributed Weather Data System | *Python, Apache Cassandra, gRPC, Protobuf, Apache Spark, Docker*

- Maintained write availability during single-node failures by designing a 3-node **Cassandra** cluster (RF=3, tunable consistency) with a **gRPC + Protobuf** ingest API and automatic read-path failover (R=2 → R=1) on **Unavailable / NoHostAvailable** exceptions, verified via `docker kill` fault injection.
- Modeled NOAA time-series records for efficient range queries by designing partition keys, clustering keys, and UDTs, then processed records with **Apache Spark** for batch analytics over the gRPC ingest path.

Real-Time Stock Streaming Engine | *Python, Apache Kafka, Protobuf, HDFS*

- Achieved **exactly-once** stock-tick processing under crash recovery by architecting a **Kafka + Protobuf** pipeline with atomic checkpointing, offset management, and consumer groups; data lands in **HDFS** for downstream analytics.
- Preserved data lineage from ingestion through transformation by isolating consumer-group offsets per stage and persisting partition state across restarts.

Multi-Source ELT Data Pipeline | *Airbyte, Snowflake, dbt, Python, SQL*

- Integrated survey data, trading-book CSVs, and stock/FX time-series into **Snowflake** by designing a multi-source ELT pipeline with **Airbyte** connectors and **dbt** transformations.
- Computed desk-level P&L across Equity and FX desks by building **6 dbt staging models** and a Snowflake fact table with NULLIF-guarded SQL division to eliminate zero-division runtime errors.

EDUCATION

University of Wisconsin–Madison

Madison, WI

B.S. Computer Sciences, Data Science, and Economics | GPA: 3.755/4.0 | Dean's List (3×)

Sep 2023 – May 2027

- Coursework:** Big Data Systems, Database Management Systems, Machine Organization & Programming, Data Structures & Algorithms, Introduction to Algorithms, Data Management for Data Science

SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Frontend Development: React, Next.js, Tailwind CSS

Backend Development: Node.js, Express.js, FastAPI, REST APIs, gRPC

Databases & Data Engineering: MongoDB, MySQL, Cassandra, Snowflake, BigQuery, Elasticsearch, dbt, Airbyte

Big Data & Distributed Systems: Apache Kafka, Apache Spark, HDFS

Cloud, DevOps & Tools: Google Cloud Platform (GCP), Docker, Git, CI/CD